

Power Outage Detection System

Using Data Mining & Machine Learning Techniques

Tetouan City Power Consumption Dataset • 52,416 Records • 3 Distribution Zones

BTech Final Year | Data Mining Course | 2025–2026

Problem & Data

- Problem Statement
- Dataset Overview
- Exploratory Data Analysis

Methodology

- Feature Engineering (39 features)
- Models: Supervised & Unsupervised
- Training Pipeline

Results & Analysis

- Architecture Tournament
- Model Comparison
- SHAP Explainability
- LSTM Autoencoder

Deployment

- Streamlit Dashboard
- Conclusion & Future Work

Why Detect Power Outages?

⚠ Real-World Impact

- Outages cost economies **\$150B+ annually**
- Hospitals & emergency services depend on uptime
- Traditional systems are **reactive** — detect after failure

💡 Our Opportunity

- 52,416 smart meter readings available
- Apply ML to **proactively** detect anomalies

🚩 Project Objectives

- 1 EDA on 52,416 records across 3 zones
- 2 Engineer 39 time-series features
- 3 Train & compare 6 ML models
- 4 Apply SHAP explainability analysis
- 5 Deploy interactive Streamlit dashboard

0.9935

XGBoost F1

1.0000

ROC-AUC

Dataset — Tetouan City Power Consumption

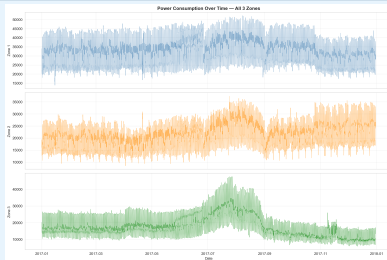


Dataset Statistics

Attribute	Value
Total Records	52,416
Date Range	Jan–Dec 2017
Frequency	10-min intervals
Zones	3 distribution
Outage Records	381 (0.727%)
Imbalance Ratio	137:1
Raw Features	8 columns
Engineered Feats	39 features

Source: UCI ML Repository — Salam & El Hibaoui (2018)

Time-Series Overview



Class Imbalance: Only 0.727% of readings are outages. Standard accuracy is misleading — **F1-score** is the correct metric.

Exploratory Data Analysis

Outage Pattern Analysis



Key Findings

- **Peak hour: 6:00 AM**
Early-morning load switching
- **Peak day: Thursday**
Mid-week operational anomaly
- **Peak month: February**
Winter demand surge
- **Outage rate: 0.727%**
Severe class imbalance
- **Zone 1 & 2: $r = -0.460$ (negative)**
- **Zone 1 & 3: $r = +0.477$ (positive)**

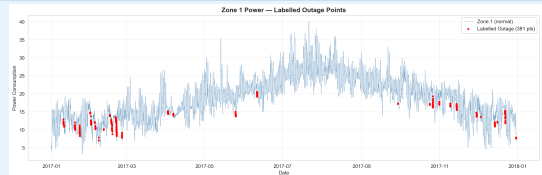
Feature Engineering — 8 Raw $\rightarrow$ 39 Features

Feature Categories

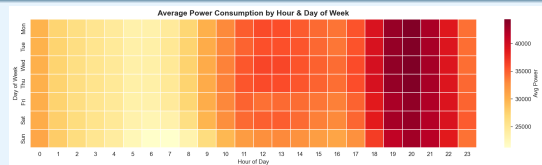
Category	Examples	N
Rolling Stats	Mean/std/min/max (2hr,24hr,48hr)	12
Lag & Diff	Lag 1/6/144 steps; diff	6
Z-Score	24hr z-score, momentum	2
Time Features	Hour, day, month, weekend, night	5
Interaction	Zone ratios, weather $\times$ power	9
Raw Features	Zone 1/2/3, temp, humidity, wind	5
Total		39

$\rightarrow$ **Top 2 SHAP features:** z1_zscore (8.456) and z1_rolling_std_w144 (0.551)

Labelled Outage Points



Outage Heatmap (Hour $\times$ Day)



Models Evaluated — 6 Models, 2 Paradigms

Supervised (with labels)

- **Logistic Regression** — linear baseline
- **Decision Tree** — rule-based baseline
- **Random Forest** — ensemble baseline
- **Neural Network** — Giant arch (1024-512-256-128-64-32), 500 epochs, cosine LR annealing, 20 training runs
- **XGBoost 🏆 Champion** — 2000-tree ceiling, early stop at round 164

Unsupervised (no labels needed)

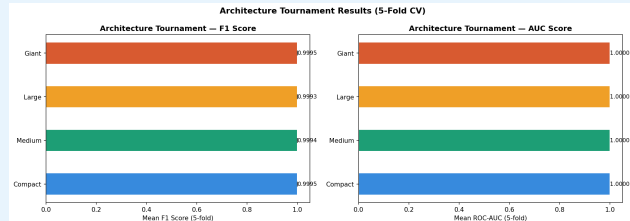
- **Isolation Forest** — random splits to isolate anomalies; 5 contamination values tuned
- **LSTM Autoencoder** — learns normal 8-hour consumption patterns; flags sequences with high reconstruction error

Why Both Paradigms?

Supervised needs labels. Unsupervised works when labelling every outage is expensive or infeasible in real deployments.

Neural Network — Architecture Tournament

Tournament Results (5-Fold CV)

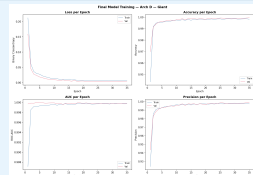


Winner: Giant Arch

1024-512-256-128-64-32

Setting	Value
Architectures	4
CV Folds	5
Training Runs	20
Max Epochs	500
Learning Rate	0.0001
LR Schedule	Cosine Anneal
Early Stop	patience=30
Epochs Ran	195

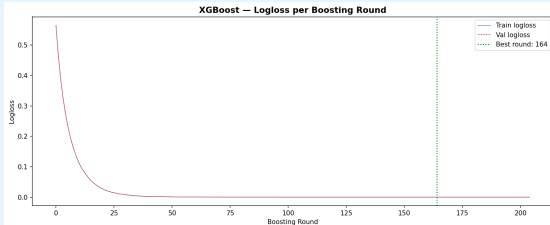
Final Model Training Curves



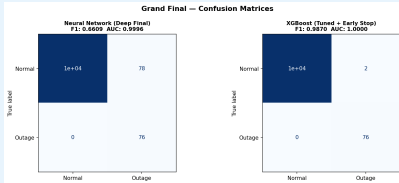
20 training runs with live epoch-by-epoch loss, accuracy and AUC logging.

XGBoost — Champion Model

Boosting Rounds — Logloss Curve



Grand Final Confusion Matrices



Training Setup

- RandomizedSearchCV:
30 combos $\times$ 5-fold = **150 fits**
- 2000-tree ceiling
early stop at round 164
- Histogram tree method
for CPU efficiency

0.9935

F1 Score

1.0000

ROC-AUC

381/381

Outages Detected

5

False Alarms

Supervised vs. Unsupervised Detection

January 2017 Detection Window — Ground Truth vs XGBoost vs Isolation Forest

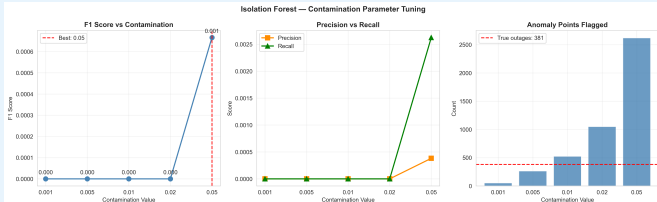


XGBoost Needs labels. Learns exact outage signatures. **F1: 0.9935**

Isolation Forest No labels needed. Point-level anomaly scores. **F1: 0.0007**

Isolation Forest — Unsupervised Baseline

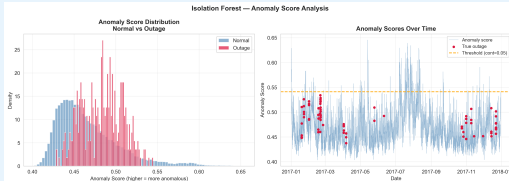
Contamination Tuning (5 values)



i How It Works

- 1 Randomly select a feature
- 2 Random split between min/max
- 3 Repeat until point is isolated
- 4 **Anomalies** = isolated in **fewer splits**
- 5 **Normal** = needs many splits

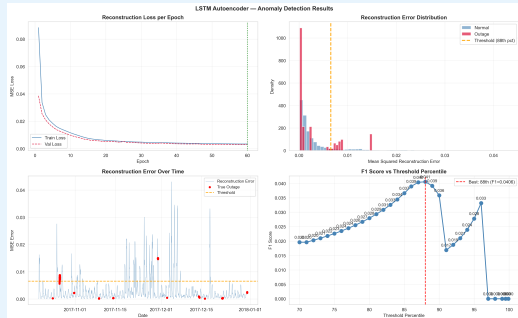
Anomaly Score Distribution & Over Time



Key insight: Low F1 is *expected* at 0.727% outage rate — outages are not statistically isolated enough for random partitioning to reliably separate them.

LSTM Autoencoder — Sequence-Level Detection

Training Loss & Detection Results



Architecture

Window: 48 steps = 8 hrs

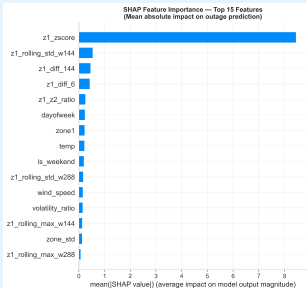
Features: Zone 1/2/3, Temp, Humidity

Input(48,5) → LSTM(64) →
LSTM(16) [Encoder] →
RepeatVector(48) →
LSTM(16) → LSTM(64) →
Dense(5) [Output]

Metric	Value
Epochs	60
Train Time	6.5 min
Best Val Loss	0.002963
F1 Score	0.0406
Recall	0.2258
Precision	0.0223

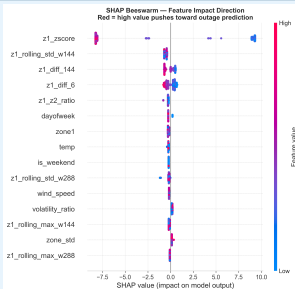
SHAP — Model Explainability

Feature Importance (Bar)



z1_zscore SHAP: 8.456 — primary anomaly signal

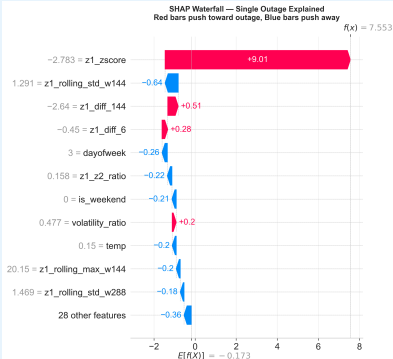
Beeswarm — Impact Direction



z1_rolling_std_w144 SHAP: 0.551 — volatility before outage

SHAP Waterfall — Single Outage Explained

Why did the model flag this reading?



Interpretation

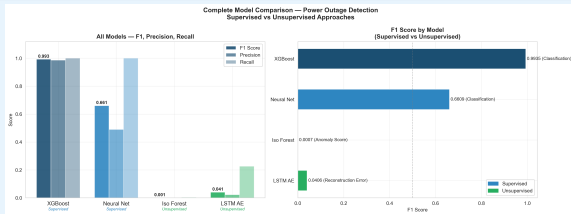
- Red bars push toward outage prediction
- Blue bars push away from outage
- $z1_zscore$ dominates — extreme deviation is the clearest signal
- SHAP = Shapley values from cooperative game theory

This slide answers: “Why did your model make this prediction?”

Results — Full Model Comparison

Model	Type	F1	Precision	Recall	AUC
XGBoost 🏆	Supervised	0.9935	0.9870	1.0000	1.0000
Neural Network	Supervised	0.9212	0.8539	1.0000	0.9999
LSTM Autoencoder	Unsupervised	0.0406	0.0223	0.2258	N/A
Isolation Forest	Unsupervised	0.0007	0.0004	0.0026	N/A

All Models F1 / Precision / Recall Comparison



Key Finding — Supervised vs. Unsupervised Gap

✗ Unsupervised Limitation

LSTM AE and Isolation Forest achieve $F1 < 0.05$ because they receive **zero label information** during training. They cannot distinguish power outages from holidays, demand spikes, or weather events.

✓ When to Use Each

- **Supervised:** labels available, production deployment
- **Unsupervised:** no labels, first-pass screening, novel anomaly discovery

🔑 Core Insight

The gap between supervised ($F1: 0.9935$) and unsupervised ($F1: 0.0406$) reveals that **label availability is the single most important factor** for outage detection on severely imbalanced time-series data.

Supervised
0.9935
vs Unsupervised
0.0406
= 24× difference in F1

Streamlit Dashboard — Live Demo

Dashboard Features

- Live time-series with detected outages highlighted in orange
- Outage probability score panel with threshold line
- Three-zone comparison with outage regions shaded in red
- **Live prediction form:** input any sensor reading, get instant probability + gauge meter
- Model switching — XGBoost, NN, Isolation Forest, LSTM AE
- Threshold slider to adjust detection sensitivity live
- Date range filter for time-window inspection

Tech Stack

- **Streamlit** — web framework
- **Plotly** — interactive charts
- **XGBoost** — champion model
- **TensorFlow/Keras** — LSTM AE
- **SHAP** — explainability
- **scikit-learn** — preprocessing

381/381

Outages Detected

5

False Alarms

across 52,416 records

Conclusion & Future Work

✔ What We Achieved

- XGBoost F1: **0.9935**, AUC: **1.0000**
- **381/381** outages detected, 5 false alarms
- **39 features** from 8 raw columns
- **20 training runs** across 4 NN architectures
- SHAP explainability for individual predictions
- 4 models across 2 paradigms compared
- Live Streamlit dashboard deployed

🔮 Future Work

- Real-time streaming with Apache Kafka / MQTT
- LSTM/Transformer for supervised sequence classification
- Multi-class outage severity prediction
- Cloud deployment with SMS / email alerting
- Federated learning across utility providers
- Extension to additional city datasets

Thank You — Questions?